

RLHF exercises

Notation

Sigmoid:

$$\sigma(x) = \frac{1}{1 + e^{-x}}.$$

Bradley–Terry preference probability:

$$P(y_1 \succ y_2 \mid x) = \sigma(r_1 - r_2).$$

Reward-model loss (pairwise logistic):

$$L = -\log \sigma(r_c - r_r).$$

PPO importance ratio:

$$\rho = \frac{\pi_{\text{new}}(a \mid s)}{\pi_{\text{old}}(a \mid s)}.$$

1 Bradley–Terry Reward Model

Exercise 1: Computing Preference Probability

Two completions have scores $r_1 = 2$ and $r_2 = 0$.

- (a) Compute $r_1 - r_2$.
- (b) Compute $P(1 \succ 2) = \sigma(r_1 - r_2)$.
- (c) Interpret the probability.

Exercise 2: Translation Invariance

Suppose $r_1 = 5$ and $r_2 = 3$.

- (a) Compute $P(1 \succ 2)$.
- (b) Add +10 to both scores and recompute the probability.
- (c) Explain what changed.

Exercise 3: Reward Model Loss

Chosen score $r_c = 1.5$ and rejected score $r_r = 2.0$.

- (a) Compute $\Delta = r_c - r_r$.
- (b) Compute $\sigma(\Delta)$.
- (c) Compute $L = -\log \sigma(\Delta)$.
- (d) Is the model doing well or poorly on this example?

Exercise 4: Gradient Direction (Conceptual)

Suppose the human prefers the chosen response and the model currently has $r_c = 0.5$, $r_r = 0.4$. Should training push r_c up, r_r down, both, or neither?

2 PPO and Advantage

Exercise 5: Simple Policy-Gradient Magnitude

A token has $\nabla \log \pi(a | s) = 0.2$ and advantage $A = 3$. Compute the proportional update magnitude $\nabla \log \pi(a | s) \cdot A$ and interpret the sign.

Exercise 6: Group Mean Baseline Advantages

For three completions of the same prompt, rewards are $[8, 5, 2]$. Use baseline = group mean.

- (a) Compute the mean reward.
- (b) Compute advantages for each reward (reward minus mean).

Exercise 7: PPO Ratio and Clipping

Old policy probability $\pi_{\text{old}} = 0.2$ and new policy probability $\pi_{\text{new}} = 0.3$ for the same token/action. Clipping range is $[0.8, 1.2]$.

- (a) Compute the ratio ρ .
- (b) Determine if it is clipped.
- (c) State the clipped value.